

Peer-to-Peer Paradigm

Napster by Shawn Fanning (1999-2001)
Gnutella, Kazaa, BitTorrent

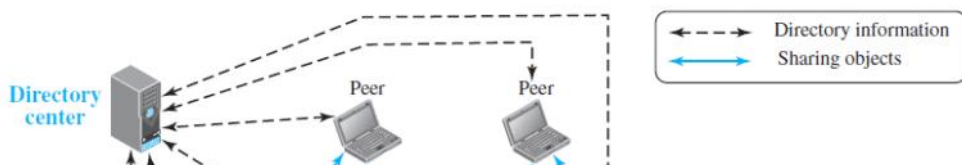
P2P Networks

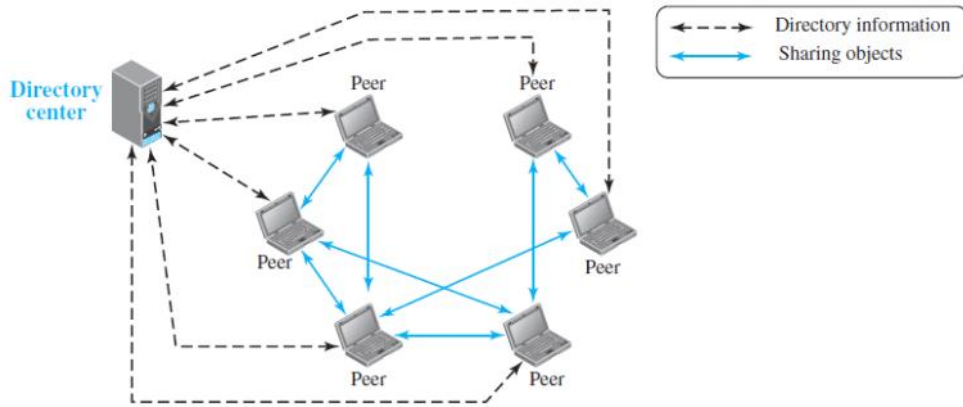
- user has a file to share
- joins a network and becomes a peer
- a peer can connect to the computer that has a file to share and download it
- multiple copies of the file become available to the group
- list of peers may grow or shrink

Q: How do we keep track of the peers? How do we keep track of the location of the files?

Centralized P2P Networks

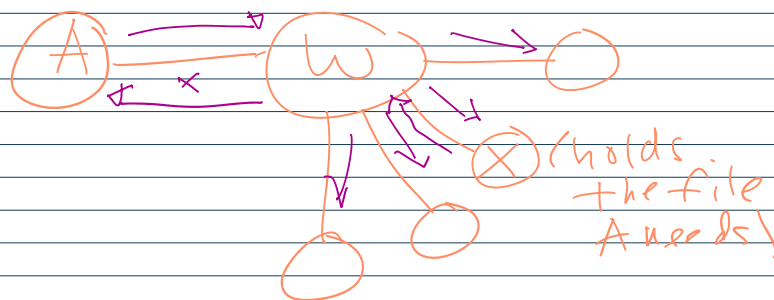
- directory system uses client-server
- actual storing and downloading uses P2P
- ex. Napster





Decentralized P2P networks

- no centralized directory system
- peers arrange themselves into an overlay network
 - ↳ a logical network on top of a physical network
- may be structured or unstructured
- ex Gnutella
 - unstructured - directory is randomly distributed between nodes



- not scalable due to flooding

flooding

- requires a peer to have at least one neighbor

→ structured networks use predefined set of rules to link the nodes so that query can be effectively resolved

used Distributed Hash Tables (DHT)

Distributed Hash Tables

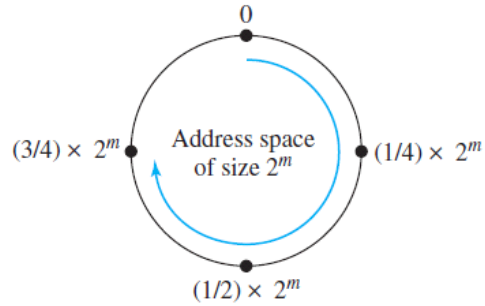
- distributes data among a set of nodes according to some predefined rules
- each peer become responsible for a range of data items
- each peer has "partial" knowledge about the whole network

address space

- data item and the responsible peer is mapped to a point in a large address 2^m
 m is typically $= 160$

Note:

1. Space range is 0 to $2^m - 1$.
2. Calculation is done modulo 2^m .



Peer identifier:

$$\text{node ID} = \text{hash}(\text{PeerIPadd}) \\ \text{C, SHA}$$

Object identifier:

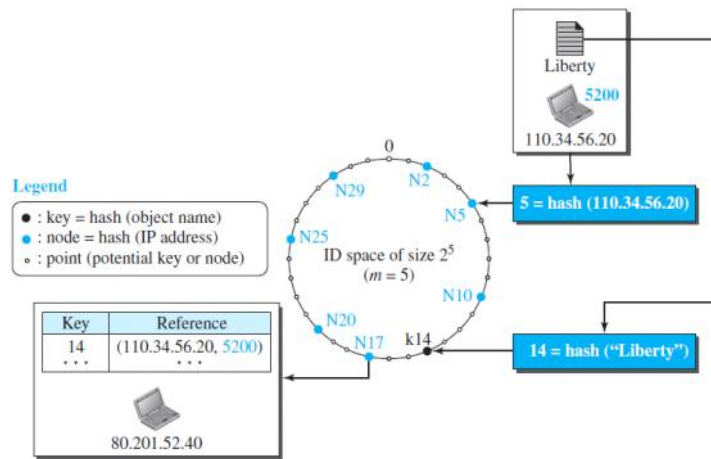
$$\text{key} = \text{hash}(\text{Object name})$$

An object in a DHT is related to the pair (key, value)

Storing an object

→ **direct method** — object is stored in the node whose id is closest to the key in the ring

→ **indirect method** — peer that owns the object keeps the object but a reference is stored in the node whose id is closest to the key in the ring



Routing

→ main function of DHT is to route a query to the node responsible for storing the reference to an object

Arrival and Departure of nodes

→ DHT implementation needs to address this. Node failure is treated as node departure

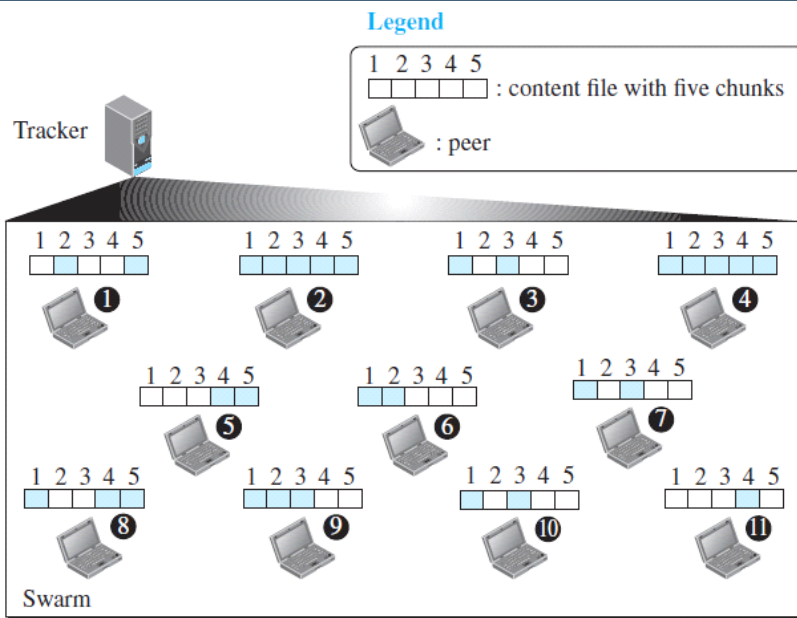
Examples of DHT implementations:
Chord, Pastry, Kademlia

BitTorrent

→ designed by Bram Cohen
→ group of peers collaborate to give all peers in the group a portion of the file ⇒ **torrent**

- to give all peers in the group a copy of the file $\Rightarrow$ **torrent**
- $\rightarrow$ **Swarm** - set of all peers that takes part in a torrent
- $\rightarrow$ **seed** - a peer that has the complete file
- $\rightarrow$ **leech** - a peer that has only part of the file

w/ Tracker



Note: Peers 2 and 4 are seeds; others are leeches.

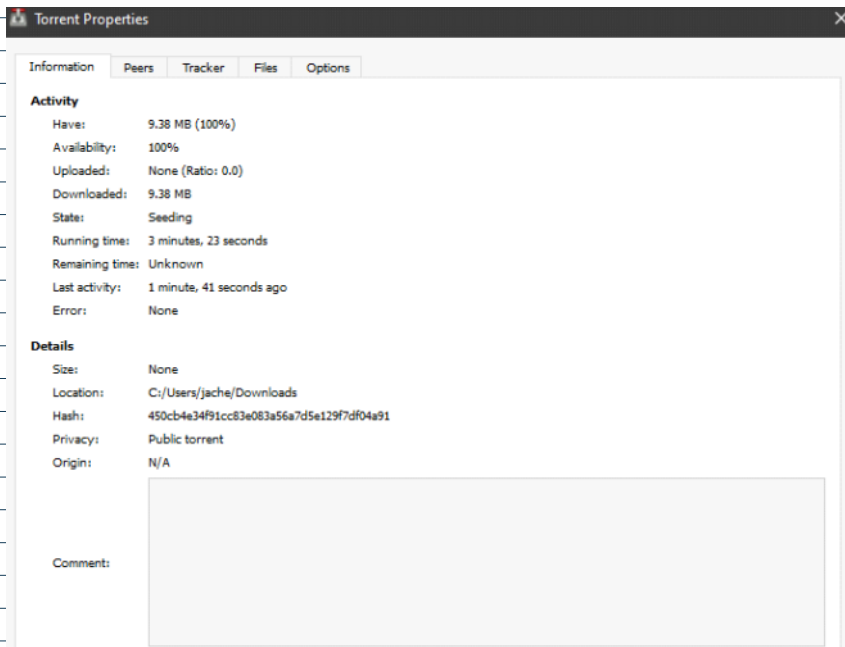
$\Rightarrow$ in trackerless, tracking is distributed using a DHT



Torrent Properties					
Information Peers Tracker Files Options					
	Up	Down	%	Status	Address Client
				T?E	45.87.250.252 Deluge 2.0.3.54
				T?E	68.34.64.104 qBittorrent 4.5.2
		84 kB/s	100%	TDE	84.117.10.181 qBittorrent 4.5.0

Information Peers Tracker Files Options	
tracker.leechers-paradise.org:6969 Got an error "Connection failed" 47 seconds ago Queued to ask for more peers	
tracker.openbittorrent.com:6969 Got a list of 8 peers 1 minute ago Asking for more peers in 25 minutes	
tracker.opentrackr.org:1337 Got a list of 9 peers 1 minute ago Asking for more peers in 28 minutes	
tracker.pirateparty.gr:6969 Got an error "Could not connect to tracker" 1 minute ago Asking for more peers in 3 minutes	
tracker.tiny-vps.com:6969 Got a list of 1 peer 1 minute ago Asking for more peers in 4 hours	
tracker.torrent.eu.org:451 Got a list of 6 peers 1 minute ago Asking for more peers in 26 minutes	

Torrent Properties	
Information Peers Tracker Files Options	
Speed ☒ Honor global limits ☐ Limit download speed: 100 kB/s ☐ Limit upload speed: 100 kB/s Torrent priority: Normal	
Seeding Limits Ratio: Use Global Settings Idle: Use Global Settings	
Peer Connections Maximum peers: 50	



The First Book of Moses, Called **Genesis** [GENESIS]

1

The Creation

- 1 In the beginning God created the heaven and the earth.
- 2 And the earth was without form, and void; and darkness *was* upon the face of the deep. And the Spirit of God moved upon the face of the waters.
- 3 ¶ And God said, Let there be light: and there was light.
- 4 And God saw the light, that *it was* good: and God divided the light from the darkness.
- 5 And God called the light Day, and the darkness he called Night. And the evening and the morning were the first day.
- 6 ¶ And God said, Let there be a firmament in the midst of the waters, and let it divide the waters from the waters.